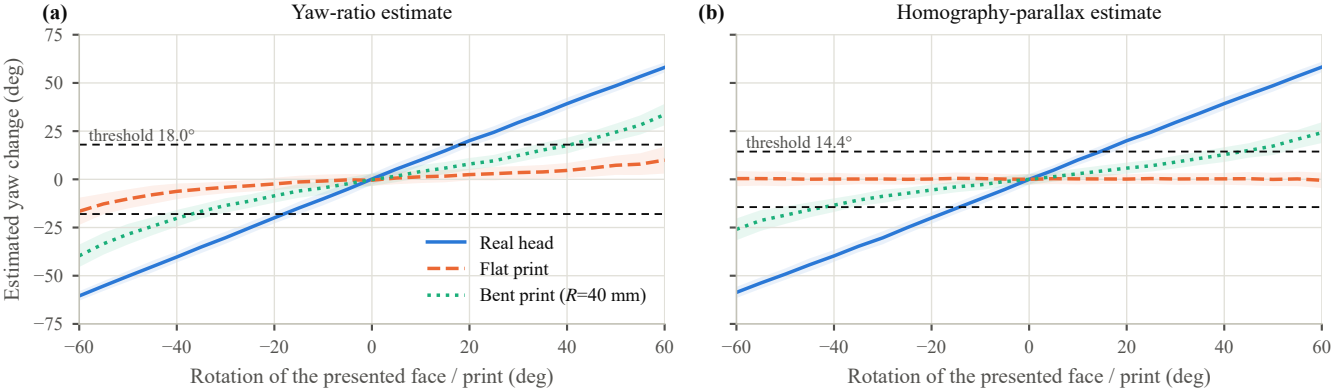

(a)
Yaw-ratio estimate
(b)
Homography-parallax estimate
Estimated yaw change (deg)
75
50
25
0
−25
−50
−75
threshold 18.0°
threshold 14.4°
Real head
Flat print
Bent print (R=40 mm)
−60
−40
−20
0
20
40
60
Rotation of the presented face / print (deg)
Rotation of the presented face / print (deg)